Unsupervised Hyperspectral Band Selection via Structure-Conserved and Neighborhood-Grouped Evolutionary Algorithm

Qijun Wang, Chaoping Song, Yanni Dong, *Senior Member, IEEE*, Fan Cheng, Lyuyang Tong, Bo Du, *Senior Member, IEEE*, and Xingyi Zhang, *Senior Member, IEEE*

***Abstract*— Hyperspectral images (HSIs) contain hundreds of bands, which provide a wealth of spectral information and enable better characterization of features. However, the excessive dimensions and redundant information also cause a dimensional disaster for subsequent processing. Band selection (BS) is a widely used dimension reduction technique for HSIs. Traditional methods mainly consider the hyperspectral BS problem at the level of data and maintain the information contained in the data, without considering the spatial structures inside HSIs. To fill the gap, in this work, an unsupervised hyperspectral BS method through structure-conserved and neighborhood-grouped evolutionary algorithm (SNEA) is proposed. Different from other EAs for hyperspectral BS, first, two spatial-structure-related optimization objectives are designed, including the locally spatial structure denoted by the pixel's spatial consistency with its adjacent neighbors and the globally spatial structure denoted by the affinity graph among pixels. With the designed objectives, the hyperspectral BS is formulated as the problem of conserving spatial structures. Moreover, a neighborhood-grouped pairwise learning strategy is proposed to generate high-quality offsprings. In this novel strategy, a neighborhood-grouping operation is developed to divide the band space into several groups. The population can be initialized efficiently, and the offspring solutions can be generated pairwisely under the guidance of grouping. Compared with nine state-of-the-art comparison algorithms, experimental results on three standard hyperspectral datasets demonstrate that the band subset obtained by our proposed SNEA has a better classification performance than the comparison algorithms.**

***Index Terms*— Band selection (BS), evolutionary algorithm (EA), hyperspectral images (HSIs), multiobjective optimization, spatial structure.**

I. Introduction

In RECENT years, hyperspectral images (HSIs) have been widely used in urban remote sensing [1], medical imaging [2], marine monitoring [3], and many other fields. However, the large number of spectral bands in HSIs brings a great challenge to the distribution, analysis, and storage [4]. Meanwhile, noisy and redundant bands in HSIs would seriously affect the accuracy of ground objects detection and classification. As a widely used dimension reduction technique for hyperspectral data, band selection (BS) is used to select a certain number of bands from all the bands and remove the redundant and noisy bands [5]. Different from feature extraction (FE) [6], which transforms the original data into the low-dimensional one by linear or nonlinear mapping, BS reserves the informative bands, whose physical meanings can be maintained and do not change the data. Therefore, hyperspectral BS gains much attention in the literature.

According to the usage of label information, hyperspectral BS can be divided into three categories: unsupervised BS, semisupervised BS, and supervised BS. Supervised BS requires many labeled samples to calculate the correlation between bands and labels and further determines the selected bands [7]. Semisupervised BS uses a small number of labeled samples and a large number of unlabeled samples to evaluate the amount of information in bands, the correlation among bands, and the degree of separation of bands from labels [8], [9]. However, the labels of samples are difficult to obtain, and in many real situations, only unsupervised BS, which does not depend on any labels, is practical. In this work, we mainly focus on unsupervised BS.

Based on the adopted strategies, the existing unsupervised BS methods consist of non-evolutionary algorithm (EA)-based methods and EA-based methods. The non-EA-based methods often include ranking-based methods, clustering-based methods, search-based methods, and hybrid methods. For ranking-based methods, different performance criteria are

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designed to evaluate the importance of each band, and the bands with best ranking (for the specific selected band number) are selected [10], [11], [12], [13]. For clustering-based methods, all the bands are divided into several clusters according to the correlation between the bands, and by selecting the bands that are closest to the centroids in each cluster, the most representative bands are obtained [14], [15], [16], [17], [18], [19], [20]. For search-based methods, the bands are sequentially selected or removed based on previously selected bands until the desired number of bands is reached [11], [21], [22]. Hybrid methods combine different types of BS approaches, such as clustering-based and ranking-based methods [10], [18], [23], to make the best use of each of them. In addition, spatial information has been widely used for HSI classification, which can dramatically improve the classification accuracy. In the hyperspectral BS, spatial information is usually used to construct the metric for ranking or clustering. Cao et al. [24] proposed the BS method based on edge map to determine the discriminating power of each band. Then, in the work of [25], spatial FE is used to reduce the dimensionality of band images and improve the BS performance. In the following, Sui et al. [26] put forward a manifold preserving (intrinsic structure) band-importance metric to measure the bandwise essentiality. The spatial information can also be merged with the spectral information to select the appropriate bands [27].

The non-EA-based methods mentioned above can solve the BS problem efficiently. However, as a typical combinatorial optimization problem, the combination of the individually best bands may not correspond to the global optimal solution. To tackle the issue, EA, as a global optimization algorithm, is utilized to handle the hyperspectral BS problem. Monteiro and Kosugi [28] first introduced particle swarm optimization (PSO) into hyperspectral BS, adopting two swarms of particles in order to simultaneously optimize a desired performance criterion and the number of selected bands, and achieved superior performance over the principal component analysis (PCA) method. After then, in recognizing the superiority of Monteiro's work, many EA-based methods [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39] were proposed and have gradually become a research hotspot in the hyperspectral BS field.

Although these EA-based methods have shown promising performance in obtaining good band subsets, these algorithms only focus on using different EA frameworks, such as PSO [28], [29], [30], [34], [36], [37], artificial bee colony (ABC) [38], [39], ant colony optimization (ACO) [35], genetic algorithm (GA) [32], [33], sine cosine algorithm (SCA) [31], and so on, to design hyperspectral BS algorithms, and the characteristics of HSIs are not fully considered. For example, HSIs are captured through remotely sensing continuous regions, which contain a large number of spatial structures. The existing methods mainly consider the hyperspectral BS problem at the level of data, without considering the spatial structure of HSIs. Meanwhile, the adjacent bands have strong correlation, and in evolutionary process, the distribution of selected bands in the band space should also be considered. To this end, in this work, an unsupervised hyperspectral BS method through structure-conserved and neighborhood-grouped EA (SNEA) is proposed to reduce the high redundancy among adjacent bands and conserve the spatial structure better. In summary, the main contributions of this article include the following three aspects.

1) Different from the previous work, the novel optimization objectives of hyperspectral BS are defined. To maintain the spatial structures inside HSIs, the optimization objectives reflecting spatial structures are designed, including local consistency with adjacent neighbors and nonlocal consistency among pixels. The former denotes the locally spatial structure, while the latter denotes the globally spatial structure. The selected band subset can be optimized to conserve spatial structures as many as possible. Then, the hyperspectral BS problem is formulated as an optimization problem of maximizing the conservation of spatial structures.
2) To solve the optimization problem of conserving spatial structures for hyperspectral BS, a neighborhood-grouping-based pairwise learning strategy is proposed to generate high-quality solutions. In this novel strategy, a neighborhood-grouping operation is designed to divide the band space into several groups. Based on the grouping, the importance of each band is measured by a distributive score. Therefore, the population can be initialized efficiently, and the offspring solutions can be generated pairwisely under the guidance of distributed scores.
3) The effectiveness of our proposed SNEA is validated on three widely used HSIs. Experimental results show that in comparison with the state-of-the-art non-EA-based and EA-based algorithms, our proposed SNEA can select the band subset with higher qualities.

The rest of this article is organized as follows. Section II introduces the related work. Then, the details of our proposed SNEA are presented in Section III. In Section IV, the experimental results of our proposed SNEA and the state-of-the-art algorithms are reported and analyzed. We conclude this article in Section V.

II. Related Work

In this section, we will review the related work to our proposed SNEA. Specifically, first, we will present the preliminaries about hyperspectral BS. Then, the work of multiobjective optimization, which is the basis of SNEA, will be introduced. Finally, we focus on the state-of-the-art and representative multiobjective EA (MOEA)-based hyperspectral BS algorithms, which are closely related to our work.

A. Preliminaries About Hyperspectral BS

Unsupervised hyperspectral BS aims to select a band subset S from the original band space R, and the band subset S should optimize the performance criterion J. Assuming the performance criterion J is optimized to get the minimum and the candidate band subset is denoted by R', the problem of hyperspectral BS can be described as follows:

$$S = \underset{R' \subseteq R}{\arg\min} \left[J\left(R'\right) \right]. \tag{1}$$

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Note that the band space R contains M bands, and the candidate band subset R' contains m bands. As BS is conducted in an unsupervised manner, the dataset D can be expressed as $\{x_i \in Z^M | 1 \le i \le N\}$, where N denotes the size of samples and Z denotes the valid sample value. For a given R', the dataset D' only reserves the sample values at the candidate band subset R' and can be expressed as $\{y_i \in Z^m | 1 \le i \le N\}$. In detail, y_i is derived from x_i by reserving the sample values at the candidate band subset R'. The performance criterion J, such as mutual information (MI) and mean variance (Var), of the candidate band subset R' can be calculated from the dataset D'.

B. Multiobjective Optimization

Multiobjective optimization involves minimizing or maximizing multiple objectives simultaneously. A general multiobjective optimization framework for the minimization problem can be expressed as follows:

$$\text{Minimize}\big(F_1(X), F_2(X), \ldots, F_{O_n}(X)\big) \tag{2}$$

where X is an M-dimensional decision vector and O_n is the number of objective functions. Given two decision vectors X_1 and X_2, if all objectives satisfy $F_i(X_1) \le F_i(X_2)$, $i = 1, 2, \ldots, O_n$, then X_1 dominates X_2, or X_2 is dominated by X_1 (denoted as $X_1 \succ X_2$). If X is not dominated by any other decision vectors, X is a Pareto optimal solution (nondominated solution). The Pareto optimal solution set is represented as Pareto front (PF) in the objective space.

In the last two decades, EA has shown to be an effective framework in solving multiobjective optimization problems (MOPs), and a lot of promising MOEAs have been developed, including non-dominated sorting genetic algorithm II (NSGA-II) [40], MOEA based on decomposition (MOEA/D) [41], indicator based evolutionary algorithm (IBEA) [42], and so on. Meanwhile, MOEAs have also been applied to solve MOPs in different applications, such as community detection [43], [44], data mining [45], vehicle routing problem [46], [47], and so on. Recently, some researchers proposed to design MOEAs to solve the hyperspectral BS problem. In Section II-C, we will review these algorithms.

C. MOEAs for Hyperspectral BS

As mentioned before, there are many EA-based methods that have shown good performance in solving hyperspectral BS problems. These EA-based algorithms include single-objective EA-based methods and MOEA-based methods. The single-objective EA-based methods aim to optimize the performance criterion, such as MI, band correlation, and mean variance [9], [28], [29], [30]. However, in these works, only one objective is considered. Compared with the single-objective EA-based methods, the MOEA-based methods could optimize multiple objectives in hyperspectral BS simultaneously without using any tradeoff parameter. Therefore, recently, the MOEA-based hyperspectral BS algorithms have been paid much more attention to.

Gong et al. [33] first introduced MOEA into hyperspectral BS, and the optimization objectives included the number of selected bands and the information entropy of bands. A multiobjective BS (MOBS) based on MOEA/D was constructed, and the optimization of each band number corresponded to a subproblem. Experimental results showed that these band subsets obtained by MOBS had a better classification performance on different datasets than non-EA-based and single-objective EA-based algorithms.

Wan et al. [31] proposed a novel discrete SCA-based multiobjective (MOSCA) feature selection approach for hyperspectral imagery to minimize the redundancy and maximize the seperability of the selected band subset. The effectiveness and universality of the proposed method were verified on five HSI datasets, and one spectral dataset of typical surface features in comparison with many single-objective EA-based methods.

Zhang et al. [38] proposed a multistrategy integrated multiobjective ABC algorithm for BS. This method could dynamically select the appropriate search operator according to the convergence and diversity of the population in the evolutionary process. Experimental results demonstrated that the proposed algorithm was a powerful approach for hyperspectral BS.

He et al. [39] proposed an unsupervised multitask ABC BS algorithm based on variable-size clustering (MBBS-VC) to simultaneously search multiple optimal band subsets with different sizes. Moreover, the intergroup collaboration strategy based on bidirectional neighborhood learning and the multimeasure integration judgment mechanism were designed to improve the performance of MBBS-VC. Compared with 15 classical BS algorithms, experimental results verified the superiority of MBBS-VC.

In summary, from the above mentioned methods, it can be noticed that MOEAs have shown better performance in solving hyperspectral BS problem in comparison with the non-EA-based algorithms and single-objective EAs. However, the existing MOEA-based hyperspectral BS algorithms mostly focus on the information at the level of data, and the characteristics of spatial and spectral structures inside HSIs are not fully considered. In our work, an unsupervised hyperspectral BS method through SNEA is proposed. In Section III, we will introduce our algorithm in detail.

III. Proposed SNEA

In this section, the details of the unsupervised hyperspectral BS through SNEA are presented. We first present the problem formulation for hyperspectral BS with an emphasis on spatial-structure conservation and then depict neighborhood-grouped pairwise learning strategy in detail. At last, we present the general framework and time complexity analysis of the proposed SNEA.

A. Problem Formulation

In traditional unsupervised BS methods, the optimization objectives usually focus on keeping the discriminant bands in classification and reducing the redundant bands, which are useless or even harmful to classification. Commonly, information entropy or mean variance of all the values at a band is used to measure the amount of information contained in a band. The MI or the correlation between bands are adopted to measure the redundancy in the band space. However, HSIs are captured

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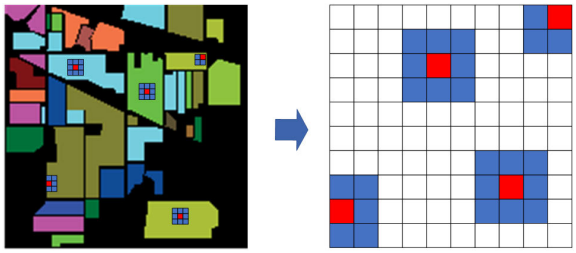


Fig. 1. Local structure is denoted by the spatial consistency with adjacent neighbors.

by remotely sensing the continuous scene, which is abundant in spatial structures, and these structures cannot be reflected by the entropy, variance, and correlation. If these metrics are taken as the optimization objectives, the spatial structure among pixels may be impaired after BS. To address this issue, the objectives reflecting spatial structures are designed, and the problem of hyperspectral BS is formulated as an optimization problem of conserving spatial structures. Before problem formulation, it is worth noting that there are many samples in an HSI, even more than 10 000. It is not possible to consider all the structures inside the HSI, and only a portion of pixels are used to construct an evaluation set (EvalSet) where spatial structures are considered.

First, it is well known that a pixel in an HSI and its adjacent neighboring pixels are likely to belong to the same class label, as shown in the left subfigure of Fig. 1. The spatial distribution of a given ground object is consistent in a continuous region. Hence, the first objective to be optimized is named a local consistency with adjacent neighbors. After BS, the difference of the current pixel from its adjacent neighboring pixels should be as less as possible. This objective can be measured by the sum of the Euclidean distances between the current pixel and its adjacent neighboring pixels and can be defined as follows:

$$f_1 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{K_l}\sum_{j=1}^{K_l}\mathrm{dist}\left(y_i, y_j\right)\right), \quad j \in N(i) \tag{3}$$

where y_i and y_j represent the pixel values on the selected bands for the ith and jth pixels in the HSI. Note that n denotes the number of sampled pixels in EvalSet, where the ith pixel belongs to, and the symbol $N(i)$ means the eight-neighboring positions of the ith pixel. Considering that, in boundary regions, the eight-neighboring positions may not all exist, then K_l denotes the number of the actual neighboring pixels around the current pixel. The function $\mathrm{dist}(y_i, y_j)$ represents the Euclidean distance between y_i and y_j. From formula (3), the local consistency objective f_1 measures the similarity between the current pixel and its adjacent neighboring pixels, reflecting the spatially local structure. This relationship should be conserved after BS. Furthermore, when f_1 is smaller, it is more likely that the current pixel and its adjacent neighboring pixels can be classified into the same category.

Since the label of each pixel is not known in advance, the Euclidean distance is not suitable for measuring the relationship between pixels that are far away from each other in an HSI. The second objective to be optimized is inspired by the intuition that the relationship between pixels should be conserved after BS, and the nonlocal consistency objective is defined as follows:

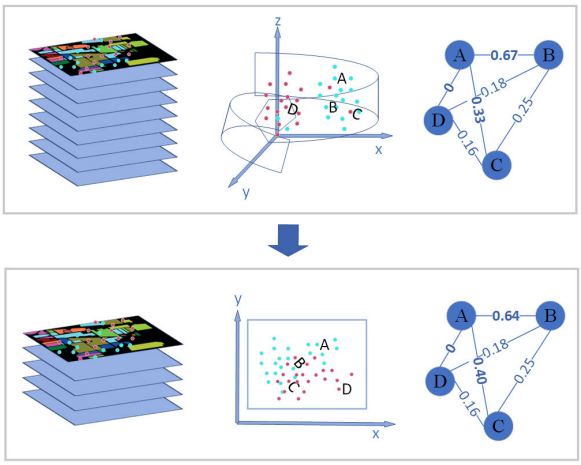


Fig. 2. Global structure is denoted by the affinity graph among pixels.

$$f_2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{n}\sum_{i'=1}^{n}S_{ii'} \times \mathrm{dist}(y_i, y_{i'})\right) \tag{4}$$

where i and i' denote two positions in EvalSet and $S_{ii'}$ denotes the affinity relationship on all bands. Considering i and i' are commonly located in departed regions, the nonlocal consistency objective f_2 actually reflects the spatially global structure. Minimizing f_2 aims at conserving the affinity relationship among pixels after BS. The affinity matrix S is constructed in the all-bands space. For two positions i and i' on all bands, the scale-invariant affinity $S_{ii'}$ is defined as follows:

$$S_{ii'} = \begin{cases} \dfrac{e_{i,K_g+1} - e_{ii'}}{K_g \times e_{i,K_g+1} - \sum_{t=1}^{K_g} e_{i,t}}, & i' \in \mathrm{EN}(i) \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where K_g represents the affinity number and e_{i,K_g+1} is the (K_g+1)th smallest Euclidean distance between x_i and the other pixels in EvalSet. Similarly, $e_{i,t}$ represents the tth smallest Euclidean distance between x_i and the other pixels in EvalSet. The Euclidean distance between x_i and $x_{i'}$ is denoted as $e_{ii'}$. $\mathrm{EN}(i)$ is the K_g nearest neighbors of the ith pixel in EvalSet under the metric of Euclidean distance.

Taking the illustration, as shown in Fig. 2, for example, in all-bands space, samples A–D are located in different positions, and the affinity relationship is shown in the graph. The Euclidean distances between A and B, C, and D are assumed to be 0.10, 0.30, and 0.50, respectively, and K_g is set to 2. By formula (5), $S_{AB} = (0.5-0.1)/((2\times 0.5)-(0.1+0.3)) = 0.67$, $S_{AC} = (0.5-0.3)/((2\times 0.5)-(0.1+0.3)) = 0.33$, and $S_{AD} = 0$ [the second branch in formula (5)]. After BS, the Euclidean distances between A and B, C, and D are assumed to be 0.11, 0.29, and 0.52, respectively. $S_{AB} = (0.52-0.11)/((2\times 0.52)-(0.11+0.29)) = 0.64$, $S_{AC} = (0.52-0.29)/((2\times 0.52)-(0.11+0.29)) = 0.40$, and $S_{AD} = 0$. Then, it can be noticed that the affinity relationship of any pair of these samples is reserved. Therefore, the BS cannot impair the global structures.

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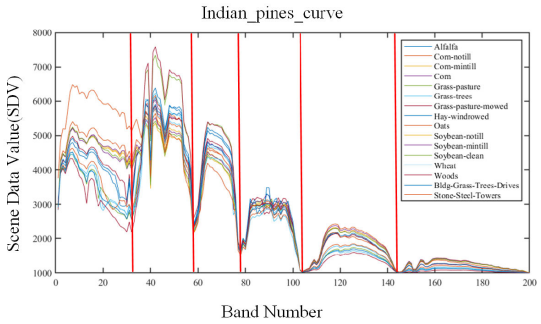


Fig. 3. Neighborhood grouping divides continuous and similar bands into a group.

In the process of hyperspectral BS, the local and nonlocal consistency objectives are optimized simultaneously to conserve the locally and globally spatial structures, and then, the problem can be formulated as follows:

$$\text{Minimize}(f_1(X),\ f_2(X)) \tag{6}$$

where X is an M-dimensional binary decision vector and 1 at a bit means the band is selected and otherwise not. The aforementioned y_i, y_j, and $y_{i'}$ can be yielded by reserving the corresponding pixel values on the selected band subset from the original values of all bands. Therefore, each decision vector X directly determines the pixel values and further influences the objective values.

For the MOP in formula (6), traditional MOEAs (e.g., NSGA-II [40] and MOEA/D [41]) can be directly utilized. However, the characteristics of HSI are not fully considered, for example, the adjacent bands have strong correlation and similar discriminating capability. Hence, the distribution of selected bands in the band space plays an important role in the classification performance. To tailor for these characteristics, a novel neighborhood-grouped pairwise learning strategy is proposed to generate high-quality offsprings.

B. Neighborhood-Grouped Pairwise Learning Strategy

The basic idea of neighborhood-grouped pairwise learning strategy lies on that the bands are divided into several continuous groups, and for each group, several representative bands are selected gradually to represent this group. This grouping strategy can maintain an even distribution for selected bands in the whole band space and avoid the congregating in some band regions. Neighborhood-grouped pairwise learning includes three key components: neighborhood grouping, importance-based pairwise learning, and solution repair, which are presented subsequently in Sections III-B1–III-B3.

1) Neighborhood Grouping: In this section, m bands are assumed to be selected. The grouping operation aims to divide the entire band space into m groups, as shown in Fig. 3. The Euclidean distance is used to measure the similarity between the bands. Note that each band is expressed as a vector, which contains all the values in the HSI at this band.

Specifically, the entire band space is uniformly divided into m groups, where the bands in a group are assigned with the same group index. Let T denote the group index vector

Algorithm 1 Neighborhood Grouping

Input : Hyperspectral image Img with N samples; Number of total bands: M; Number of selected bands: m
Output: Grouping result T

```
for j = 1 to M do
    B(j) ← Concatenate all the values at j-th band in Img;
    R(j) ← ∞;
end
C(0) ← 1;
for t = 1 to m do
    C(t) ← t * M/m;
    T(t * M/m + 1 : (t + 1) * M/m) ← t;
end
C(m + 1) ← M ;
for t = 1 to m do
    for j = C(t − 1) to C(t + 1) do
        D(j) ← dist(B(j), B(C(t)));
        if D(j) < R(j) then
            R(j) ← D(j) ;
            T(j) ← t;
        end
    end
end
return T;
```

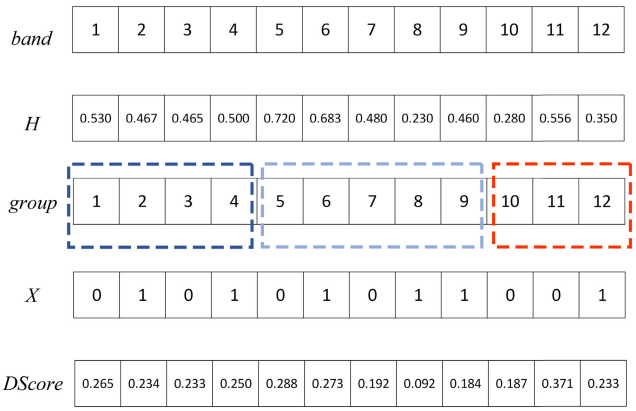


Fig. 4. Example for the neighborhood grouping and the following calculation of distributive score.

for each band. Assuming the current group is t satisfying $1 \le t \le m$, the initial group center for this group is C_t. For each band, its similarity with its two adjacent group centers is compared, and this band belongs to the group whose group center has a higher similarity with the current band. Finally the grouping result T can be obtained. The detailed procedure of neighborhood grouping can be referred to Algorithm 1.

Given the grouping result of the entire band space, there are at least two benefits. First, the population can be initialized efficiently. A band is randomly selected from the bands in each group, and then, an individual containing m selected bands can be formed. The selected bands in this individual are evenly distributed in the band space. This initialization procedure

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is named group-based individual initialization hereafter and will be utilized in the following algorithms. Second, the importance of each band in a group is influenced by the size of selected bands in the group. If more bands are selected in a group, the importance of bands in the group will be decreased to maintain an even distribution in the band space and vice versa. Commonly, the entropy at a band denotes the information amount contained in this band and can be calculated from the distribution histogram of the values at this band in the HSI. Combining with the size of selected bands in a group, a distributive score (DScore) is designed to measure the importance of a band and is defined as follows:

$$\text{DScore}(i) = \frac{H_{\max} - H_i}{H_{\max} - H_{\min}} \times \left(1 - \frac{N_{tx}}{N_t}\right) \tag{7}$$

where N_{tx} represents the size of selected bands in the group t and N_t represents the total number of bands in the group t. H_i represents the information entropy of the ith band. $H_{\max}$ and $H_{\min}$ represent the maximum and minimum of information entropy among all bands. The normalization through maximum and minimum values is used to restrict the entropy into the scope of [0, 1], ensuring that the distributive score is located in [0, 1].

Taking the illustration, as shown in Fig. 4, for example, there are 12 bands whose information entropy is assumed to be the values in the following row. After neighborhood grouping (Algorithm 1), the bands can be divided into three groups: 1–4, 5–9, and 10–12. Therefore, $N_1 = 4$, $N_2 = 5$, and $N_3 = 3$. For an individual X (the row indicated by X in Fig. 4), the selected band indices for each group are (2, 4), (6, 8, 9), and (12), and $N_{1x} = 2$, $N_{2x} = 3$, and $N_{3x} = 1$. Through formula (7), DScore can be obtained, e.g., $\text{DScore}(1) = 0.530 \times (1-2/4) = 0.265$, $\text{DScore}(2) = 0.467 \times (1 - 2/4) = 0.234$, and $\text{DScore}(6) = 0.683 \times (1 - 3/5) = 0.273$.

2) Importance-Based Pairwise Learning Strategy: After the neighborhood-grouping process, the band space is divided into m groups. In our work, the population is initialized based on the unit of band group instead of random initialization. After initialization, an importance-based pairwise learning strategy is adopted to generate solutions. In the importance-based pairwise learning strategy, two individuals are chosen randomly at each time and updated under the guidance of the distributive scores of all bands in each individual. For the two randomly selected individuals, the shift-based density estimation (SDE) [48] is used to discriminate which individual is good (lower SDE value) or bad. The good individual is denoted by X_g, while the bad individual is denoted by X_b hereafter.

In importance-based pairwise learning strategy, the bad individual X_b learns from the good one X_g. Obviously, this pairwise learning process includes two crucial issues: 1) which part of X_g can be learned by X_b and 2) how to learn from X_g. Considering the individual is represented with binary encoding scheme, the update operation on each bit of X_b is actually a binary flipping, which includes 0–1 flipping and 1–0 flipping. For the first issue, the different parts of X_g from X_b are all considered as the learning targets. Then, the difference of X_g from X_b can be obtained by

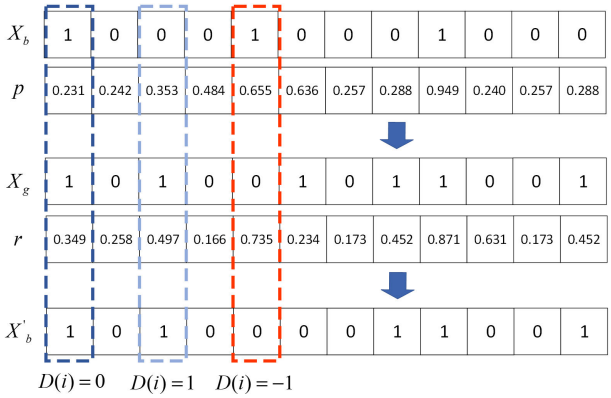


Fig. 5. Example for the learning of a bad individual from a good one.

$$D(i) = X_b(i) - X_g(i) \tag{8}$$

where i denotes the band index. If $D(i)$ is equal to 0, it means that at the ith bit X_b and X_g are the same, and X_b is not necessary to learn from X_g at this band. When $D(i)$ is equal to 1, it means the value of X_b at the ith bit is 1, and the value of X_g at the ith bit is 0, and for X_b, it should be determined whether the ith bit of X_b is flipped to 0. When $D(i)$ is equal to -1, it means the value of X_b at the ith bit is 0, and the value of X_g at the ith bit is 1, and for X_b, it should be determined whether the ith bit is flipped to 1. In our work, a probability $p(i)$, which is used to control the bit flipping of the ith bit in X_b, is designed

$$p(i) = \begin{cases} cr_i \times D(i) \times \text{DScore}(i), & \text{if } D(i) = 1 \\ -cr_i \times D(i) \times (1 - \text{DScore}(i)), & \text{if } D(i) = -1 \end{cases} \tag{9}$$

where $\text{DScore}(i)$ is the distributive score of the ith band in X_b. For X_b, DScore can be directly mapped to the flipping probability. To maintain the diversity, a random variable cr_i located in [0, 1] is further introduced to generate the ultimate flipping probability. The probability contains two cases. When $D(i)$ is equal to 1, if the random variable r_i is less than $p(i)$, the 1 at the ith bit of X_b is flipped to 0. When $D(i)$ is equal to -1, if the random variable r_i is less than $p(i)$, the 0 at the ith bit of X_b is flipped to 1.

Taking the illustration, as shown in Fig. 5, for example, two individuals are selected and compared under the metric of SDE values; then, X_g and X_b are determined. X_b has three selected bands, and the flipping probability $p(i)$ is calculated through (9). The X_g has five selected bands. A serial of random numbers are generated and shown in the row indicated by r. Therefore, when $D(i)$ is equal to 0, such as the first bit, $X(1)$ remains unchanged. When $D(i)$ is equal to 1 and r_i is less than $p(i)$, such as the third bit, $X(3)$ is flipped to 1. When $D(i)$ is equal to -1 and r_i is less than $p(i)$, such as the fifth bit, $X(5)$ is flipped to 0. Then, an offspring individual X'_b is generated.

Besides, X_g is updated based on a disturbance strategy. The basic idea lies on the following three aspects: first,

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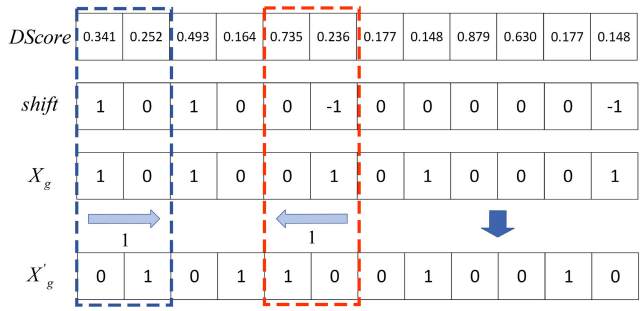


Fig. 6. Example for the update on the good individual based on a disturbance strategy guided by distributive score.

once the importance (DScore) of each band in an individual is determined, the selected unimportant band has a larger probability to shift to its adjacent bands, while the selected important band has a smaller probability to shift. Second, the shifting step is set to a larger value at the start and becomes smaller along the evolution process. Third, the shifting is also related to the size of selected bands. If m is large, the shifting can be advanced with a smaller step, while if m is small, the shifting step can be large. Combining all these considerations together, the shifting step can be defined as follows:

$$\text{shift}_i = \lceil r_{\text{shift}} \times (1 - \text{DScore}(i)) \rceil. \tag{10}$$

Note that r_{shift} is a random variable uniformly sampled from the range $[-\text{range}, \text{range}]$, and the range is related to m and the number of iterations

$$\text{range} = \left\lceil \frac{\text{maxIt} - \text{it}}{\text{maxIt}} \times \frac{m_{\max} - m}{m_{\max} - m_{\min}} \times \frac{M}{100} \right\rceil \tag{11}$$

where maxIt denotes the maximum number of evolutionary iterations and it denotes the current number of iterations. $m_{\max}$ and $m_{\min}$ denote the maximum number and minimum number of selected bands required in hyperspectral BS.

Taking the illustration, as shown in Fig. 6, for example, X_g has four selected bands, and its distributive score is located in the first row. In the update process, the shift step is calculated through formulas (10) and (11). Note that shift_1 is 1, and shift_6 is -1. At last, the selected band shifts 1 to the right at the first band and shifts 1 to the left at the sixth band. Then, an offspring individual X'_g is generated.

Specifically, the importance-based pairwise learning strategy is presented in Algorithm 2.

3) Solution Repair: For the new individual X generated by the above importance-based pairwise learning strategy, there may be a situation that the size of selected bands is greater or less than the preset number of bands m. Therefore, the offspring X should be repaired, so that the number of selected bands can be equal to m. First, for the individual X that needs to be repaired, the selected subset (bs) and the unselected band subsets (bs') are obtained from X, and the size of selected band subset is denoted by bn. If the selected number of bands bn is greater than m, $(bn - m)$ bands need to be removed from bs. In this situation, the information entropy of each band in bs is sorted in an ascending order, and the $(bn - m)$ bands with the smallest information entropy are removed from bs. Similarly, if the number of selected bands bn is less than m, then $(m - bn)$ bands in bs' need to be added to bs. Then, the information entropy of each band in bs' is sorted in a descending order, and the $(m - bn)$ bands with the largest information entropy in bs' are added into bs.

Algorithm 2 Importance-Based Pairwise Learning

Input : Good individual X_g and its distributive score $DScore_g$; Bad individual X_b and its distributive score $DScore_b$; Number of total bands M

Output: the offsprings X'_b and X'_g

1 $X'_b \leftarrow X_b$, $X'_g \leftarrow X_g$;
2 **for** $j = 1$ **to** M **do**
3 $P(j) \leftarrow$ Calculate the flipping probability by (10) from $DScore_b$;
4 $D(j) \leftarrow$ Calculate the pair difference by (9);
5 **if** $D(j) == 1$ *and rand()*$< p(j)$ **then**
6 $X'_b(j) = 0$;
7 **end**
8 **if** $D(j) == -1$ *and rand()*$< p(j)$ **then**
9 $X'_b(j) = 1$;
10 **end**
11 **end**
12 **for** $j = 1$ **to** M **do**
13 **if** $X_g(j) == 1$ **then**
14 $shift_j \leftarrow$ Calculate shift by (11) from $DScore_g$;
15 $X'_g(j) \leftarrow 0$;
16 $X'_g(j + shift_j) \leftarrow 1$;
17 **end**
18 **end**
19 return X'_b and X'_g;

In addition, during the evolutionary process, there will be a situation that multiple bands of a group are selected, while there are no bands selected in some groups. In this situation, the distribution in the band space for the individual is not good, and a threshold is introduced, so that if the number of groups with no bands exceeds the threshold, this individual is reinitialized based on the neighborhood grouping.

Specifically, the solution repair strategy is presented in Algorithm 3.

C. General Framework

With the suggested components above, we will introduce the general framework of our proposed SNEA, whose detailed procedure is presented in Algorithm 4.

The general framework of SNEA can be divided into three stages: the initialization stage, the main-loop stage, and the output stage. For the initialization stage (Lines 1–4), the EvalSet is constructed, and the groups of the band space are obtained through neighborhood grouping. According to the group division, the population P can be initialized (see the third paragraph in Section III-B1). Afterward, the distributive score of each band in an individual can be obtained. Then, our proposed SNEA comes to the main-loop stage (Lines 5–23). Obviously, the main-loop stage consists of the outer and inner

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Algorithm 3 Solution Repair

Input : The individual to be repaired X; Size of selected bands m; Number of the total bands M; Grouping result T
Output: Individual after repair X'

```
1  bs, bs' ← Get the selected and unselected band subsets from X;
2  bn ← length(bs);
3  H(1 : M) ← Calculate the band entropy of all bands;
4  if bn > m then
5  |  Remove bn − m bands with the lowest band entropy in bs;
6  |  X' ← Convert bs to an individual;
7  else
8  |  Add m − bn bands with the highest band entropy in bs' into bs;
9  |  X' ← Convert bs to an individual;
10 end
11 gn ←Count the number of groups with no selected band in X';
12 if gn > Th then
13 |  X' ←Group-based individual initialization (T);
14 end
15 return X';
```

Algorithm 4 General Framework of SNEA

Input : Hyperspectral image Img with N pixels and M bands; Number of selected bands m; Population size NP
Output: Band subset S

```
1  EvalSet ← Randomly select n samples from Img;
2  T ←Neighborhood Grouping(Img, N, M, m); // see Algorithm 1
3  P ←Group-based individual initialization(T) by NP times;
4  P' ← ∅;
5  for it = 1 to maxIt do
6  |  while |P| > 1 do
7  |  |  {p, q} ← Randomly select two individuals from P;
8  |  |  P ← P \ {p, q};
9  |  |  if SDE(p) < SDE(q) then
10 |  |  |  X_g ← p, X_b ← q;
11 |  |  else
12 |  |  |  X_g ← q, X_b ← p;
13 |  |  end
14 |  |  DScore_b, DScore_g ←Calculate the distributive score for X_b and X_g by (7);
15 |  |  X'_b, X'_g ←Importance-Based Pairwise Learning (X_b,X_g,DScore_b,DScore_g,M);
16 |  |  // see Algorithm 2
17 |  |  X'_g ←Solution Repair (X'_g, M, m, T);
18 |  |  // see Algorithm 3
19 |  |  X'_b ←Solution Repair(X'_b, M, m, T);
20 |  |  P' ← P' ∪ {X_g, X_b, X'_g, X'_b};
21 |  end
22 |  P ← Environmental Selection (P');
23 end
24 X_best ←Select the best individual from P ;
25 S ←Convert X_best into the band subset;
26 return S;
```

loops and is the main part of the whole procedure. For the inner loop, two individuals are randomly selected from P, and the good individual and bad individual are determined. Then, importance-based pairwise learning is conducted to generate offspring individuals. The bad individual learns useful part from the good individual, while the good individual is imposed on disturbance-based solution update. After then, the solution repair is adopted to modify the generated two offspring individuals. To maintain the diversity of P as much as possible, the two parent individuals are excluded from P, while the two offspring individuals and the two parent individuals are inserted into another population P'. Then, the inner loop continues if the criterion is satisfied (the size of P is larger than 1). When the inner loop is finished, environmental selection is utilized to select the updated P, and the outer loop is continued if the criterion is satisfied (the iteration number is smaller than maxIt). For the output stage (Lines 24–26), the best individual is selected from P according to the metrics, such as f_1, f_2, MI, Var, SDE, and so on, whose results will be discussed in Section IV. At last, the band subset is obtained from the best individual and returned.

D. Computational Complexity Analysis

According to the general framework of SNEA (see Algorithm 4), the SNEA algorithm mainly includes three stages.

1) *Initialization Stage (Lines 1–4):* In this stage, the EvalSet containing n pixels is constructed $[O(n)]$; the whole band space including M bands is divided into m groups $[O(m \times M)]$, and the population including NP individuals is initialized $[O(M \times NP)]$. Thus, the total time complexity of the initialization stage is $O(n + m \times M + M \times NP)$.
2) *Main-Loop Stage (Lines 5–23):* The main-loop stage includes the outer loop for each iteration generation and the inner loop for each pair of individuals in the population. For the maxIt generations and the population size NP, Lines 7–13 are used to determine the good and bad individuals for each pair, and their time complexity is $O(\text{maxIt} \times NP)$. The computational complexity of SNEA is determined primarily by four operations: the calculation of the distributive score (Line 14), importance-based pairwise learning (Line 15), solution repair (Lines 16 and 17), and environmental selection (Line 22). As mentioned in Algorithms 1–3, the calculation of the distributive score, importance-based pairwise learning, and solution repair are all conducted in a bitwise (bandwise) way, and the time complexity for these three operations is $O(\text{maxIt} \times NP \times M)$. For environmental selection, for every generation, the number of imported solutions for environmental selection is

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$2 \times NP$. NP novel individuals need to be evaluated, and the time complexity is $O(\text{maxIt} \times NP \times n)$. The time complexity of nondominated sorting in environmental selection is $O(\text{maxIt} \times NP^2)$. Thus, the time complexity of environmental selection is $O(\text{maxIt} \times NP^2) + O(\text{maxIt} \times NP \times n)$. The time complexity of the main-loop stage is the sum of all these operations: $O(\text{maxIt} \times NP) + O(\text{maxIt} \times NP \times M) + O(\text{maxIt} \times NP^2) + O(\text{maxIt} \times NP \times n) = O(\text{maxIt} \times NP \times (M + NP + n))$.

3) *Output Stage (Lines 24 and 25):* In this stage, the best solution is selected and converted into the band subset, and the time complexity is $O(NP + M)$. As $m \ll \text{maxIt} \times NP$, the total time complexity of our proposed SNEA is equal to $O(\text{maxIt} \times NP \times (M + NP + n))$.

IV. Experimental Results

In this section, a series of experiments are conducted to verify the effectiveness of SNEA. Specifically, we first present the experimental settings, which include comparison algorithms, benchmark datasets, and the parameter settings for all the algorithms. Then, the experimental results are reported and analyzed in detail. Third, we validate the effectiveness of the suggested strategies in SNEA. In the following, the parameter sensitivity analysis of the proposed SNEA is presented. Afterward, the performance of SNEA on the challenging Houston dataset and with the CNN-based classification model is reported subsequently. Finally, the running time of all the algorithms is presented in the last part.

A. Experimental Settings

1) Comparison Algorithm: To verify the performance of our proposed SNEA, nine state-of-the-art hyperspectral BS algorithms are adopted as the comparison algorithms, including five up-to-date non-EA-based unsupervised hyperspectral BS methods (namely, fast neighborhood grouping method for hyperspectral band selection (FNGBS), ONRBS, local-view-assisted discriminative band selection with hypergraph autolearning for hyperspectral image classification (LvaHAI), adaptive distance-based band hierarchy for effective hyperspectral band selection (ADBH), and hyperbolic clustering-based band hierarchy for effective unsupervised band selection (HCBH), respectively) and four up-to-date EA-based algorithms containing one single-objective EA-based method [spectral spatial GA-based unsupervised band selection (SSGA)] and three MOEA-based methods (MBBS-VC, MOBS, and MOSCA). The reason for choosing them as the comparison algorithms lies in the fact that all of them are the recently suggested hyperspectral BS algorithms, which have shown better performance than many other BS methods.

1) *FNGBS [14]:* The HSI cube in space is partitioned into several groups using a coarse-to-fine strategy. The bands with the maximum product of local density and information entropy in groups are selected as band subset.
2) *Optimal Neighborhood Reconstruction Based Band Selection (ONRBS) [21]:* Hyperspectral BS is considered as a combinatorial optimization problem. A band combination is evaluated by assessing its ability to reconstruct the original data, and a noise reducer is applied to minimize the influence of noisy bands.
3) *LvaHAI [13]:* The whole band space is randomly divided into several subspaces of different dimensions, and a robust hinge loss function for isolated pixels regularized by the row sparsity is adopted to measure the importance of the corresponding bands. Finally, a hypergraph model that automatically learns the hyperedge weights is constructed to select bands.
4) *ADBH [19]:* An adaptive distance-based band hierarchy clustering framework is designed for unsupervised BS, and this method can help to avoid the noisy bands while reflecting the hierarchical data structure of HSI.
5) *HCBH [20]:* With a tree-based hierarchy in the hyperbolic space, HCBH can effectively reveal the geometrical structure of the HSI dataset. By introducing a cluster-center-based ranking metric, the desired band subset can be naturally obtained during the clustering.
6) *SSGA [49]:* In SSGA, superpixel segmentation is used to divide the whole image into several superpixels. The GA algorithm is adopted to find the optimal band subset with the optimal fisher score.
7) *MBBS-VC [39]:* The hyperspectral BS problems with different band numbers are modeled into several subtasks, and a multitask framework with the artificial swarm algorithm is used to find band subsets.
8) *MOBS [33]:* In MOBS, the number of targets and the information amount of bands are optimized simultaneously, and a multiobjective optimization framework based on MOEA/D is constructed to obtain several optimal nondominated solutions.
9) *MOSCA [31]:* The hyperspectral BS is formulated as an optimization problem on the maximal information and minimal redundancy, simultaneously. A discrete sine and cosine algorithm is constructed to solve this MOP.

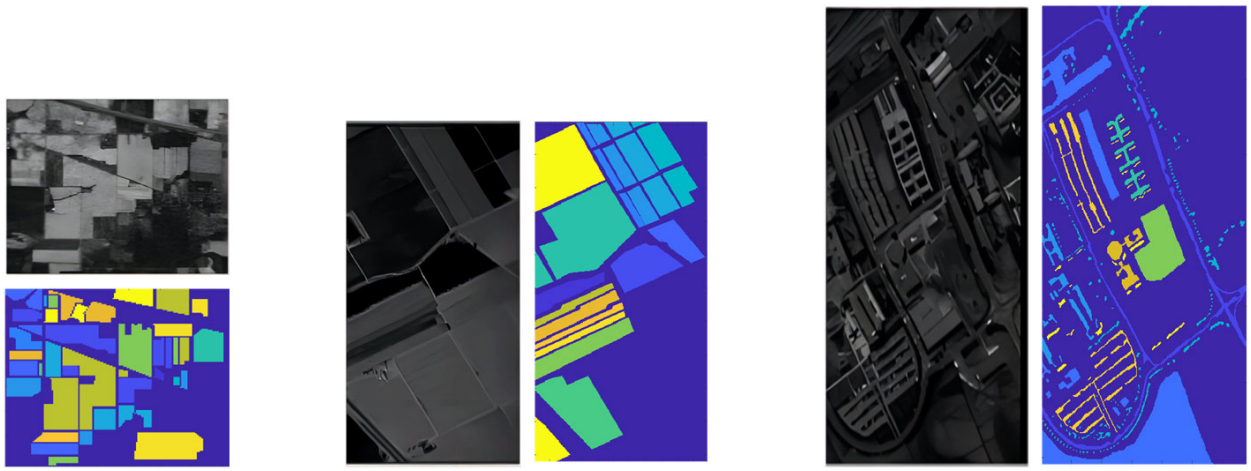
2) Datasets: Three classic HSIs (Indian_Pines, Salinas, and PaviaU), which have been widely used in evaluating the performance of different hyperspectral BS algorithms, are adopted as the benchmark datasets and can be downloaded from the website.[1]

Indian_Pines consists of 145×145 pixels and 220 spectral bands, and 20 spectral bands in 104–108, 150–163, and 200 are removed due to water absorption. The Indian_Pines scene contains 16 classes of different ground objects, and 10 249 pixels are labeled. Salinas is characterized with 512×217 pixels and 224 spectral bands, and 20 bands in 108–112, 154–167, and 224 are removed due to water absorption. The Salinas scene contains 16 classes of different ground objects, and 54 129 pixels are labeled. PaviaU has 610×340 pixels and 103 spectral bands. The PaviaU scene has nine classes of different ground objects, and 42 776 pixels are labeled. The three HSIs and the ground truth are shown in Fig. 7, and the detailed classes and the sample number of each class in these 3 datasets are presented in Table I.

3) Parameter Settings: Similar to other hyperspectral BS algorithms, the overall accuracy (OA), Kappa coefficient (Kappa), and average accuracy (AA) of the classification

[1] https://rslab.ut.ac.ir/data

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(a) (b) (c)

Fig. 7. Sample band and ground truth of the datasets (a) Indian_Pines, (b) Salinas, and (c) PaviaU.

TABLE I

GROUND TRUTH CLASSES OF INDIAN_PINES, SALINAS, PAVIAU, AND THEIR SAMPLE NUMBERS

Indian_pines			Salinas			PaviaU		
#	Class	Samples	#	Class	Samples	#	Class	Samples
1	Alfalfa	46	1	Brocoli_green_weeds_1	2009	1	Asphalt	6631
2	Corn-notill	1428	2	Brocoli_green_weeds_2	3726	2	Meadows	18649
3	Corn-mintill	830	3	Fallow	1976	3	Gravel	2099
4	Corn	237	4	Fallow_rough_plow	1394	4	Trees	3064
5	Grass-pasture	483	5	Fallow_smooth	2678	5	Painted metal sheets	1345
6	Grass-trees	730	6	Stubble	3959	6	Bare Soil	5029
7	Grass-pasture-mowed	28	7	Celery	3579	7	Bitumen	1330
8	Hay-windrowed	478	8	Grapes_untrained	11271	8	Self-Blocking Bricks	3682
9	Oats	20	9	Soil_vinyard_develop	6203	9	Shadows	947
10	Soybean-notill	972	10	Corn_senesced_green_weeds	3278	#	total-samples	42776
11	Soybean-mintill	2455	11	Lettuce_romaine_4wk	1068			
12	Soybean-clean	593	12	Lettuce_romaine_5wk	1927			
13	Wheat	205	13	Lettuce_romaine_6wk	916			
14	Woods	1265	14	Lettuce_romaine_7wk	1070			
15	Buildings-Grass-Trees-Drives	386	15	Vinyard_untrained	7268			
16	Stone-Steel-Towers	93	16	Vinyard_vertical_trellis	1807			
#	total-samples	10249	#	total-samples	54129			

on testing data are taken as the evaluation metrics. OA is the ratio of correctly classified pixels over the total pixels. AA is the average value of the classification accuracy on all classes. Kappa represents an indicator for consistency checking. Two widely used classifiers K nearest neighbors (KNNs) and support vector machine (SVM) are adopted to evaluate the performance of the selected band subsets obtained by different algorithms. For the KNN classifier, the nearest neighbor number is set as $K = 5$. For the SVM classifier, the radial basis function (RBF) is taken as the kernel function, where c is set to be 5000 and g is set to be 0.1. For each dataset, 10% samples of each class are randomly selected as the training data, and the remaining samples are used as the testing data. After a band subset is obtained by a hyperspectral BS algorithm, the pixel values of the selected bands are taken as the input for KNN and SVM classifiers, and the OA, AA, and Kappa are collected on the testing data. To achieve more stable results. The OA, AA, and Kappa results are the average values of ten trials. For the minimum of the selected bands, number m is set to 4, while the maximum is set to 60. In another word, for every m value, each algorithm is executed once to get its band subset, whose classification results are the average values of ten trials.

To ensure the comparison fairness, we adopt the recommended parameters values for all the comparison algorithms, which were suggested in their original papers. For SNEA, EvalSet is constructed by randomly selecting 10% of the total samples if the dataset size is less than 30 000 and selecting 5% of the total samples if the dataset size is more than 30 000. The population size and iteration times are set to 50 and 50, respectively. The reinitialized threshold Th in Algorithm 3 is equal to $m \times m/60$, where 60 is the maximum size of selected bands, and Th is an adaptive threshold with the increase of m.

B. Overall Performance

In this section, the results of all the algorithms on each dataset are reported and analyzed, respectively.

1) Indian_Pines: On the Indian_Pines dataset, the different band subsets can be obtained through all the algorithms, and then, the performance of these band subsets is evaluated with the KNN and SVM classifiers, respectively. The OA,

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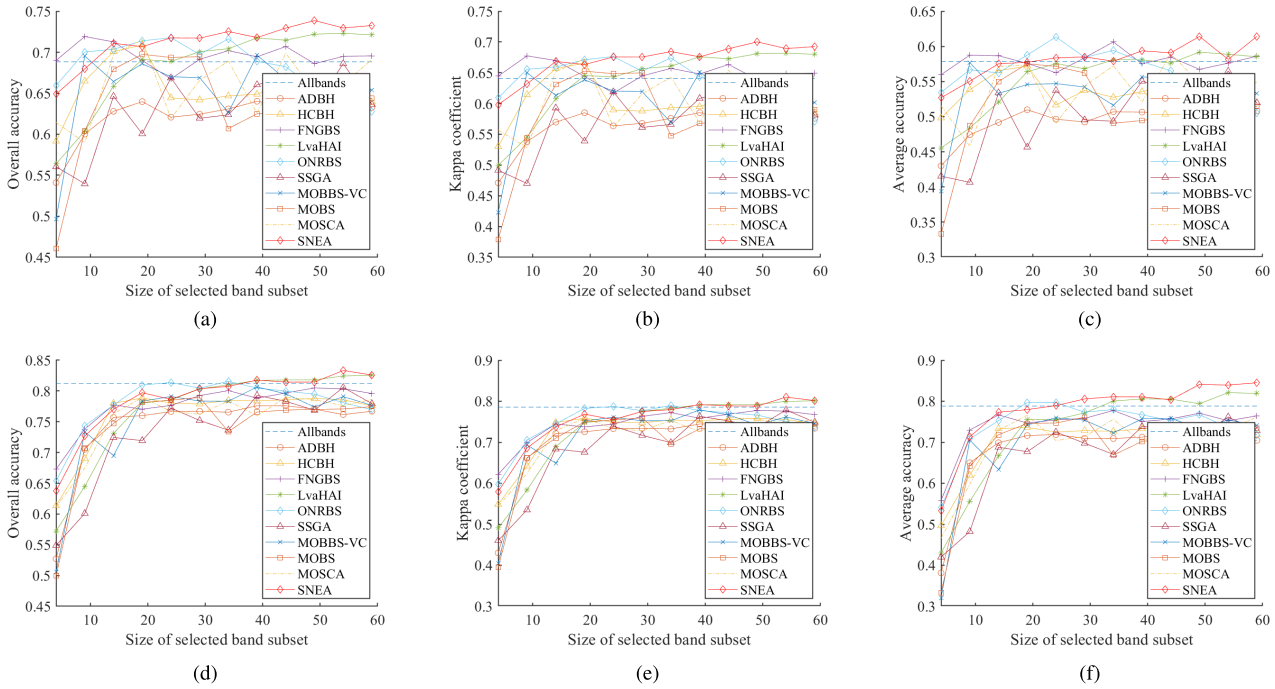


Fig. 8. OA, Kappa, and AA results of all the algorithms on the Indian_Pines dataset with the KNN classifier and the SVM classifier. (a) KNN-OA. (b) KNN-Kappa. (c) KNN-AA. (d) SVM-OA. (e) SVM-Kappa. (f) SVM-AA.

Kappa, and AA results of the KNN and SVM classifiers on the Indian_Pines dataset are illustrated in Fig. 8, where the x-axis represents the size of selected bands and the y-axis indicates OA, Kappa, and AA, respectively. The blue-dashed and horizontal line denotes the result of all bands. For the KNN classifier, it can be noticed that when the size of selected bands is small (4–10), the accuracy of most of the algorithms is less stable. When the size of selected bands is more than 10, the performance of our proposed SNEA becomes stable and is significantly better than other algorithms. Note that the accuracy of the selected bands obtained from our proposed SNEA is even higher than that of all bands, which means that the selected bands are more effective, and the noisy and redundant bands bringing negative effect on the classification performance can be removed through our proposed SNEA. For the SVM classifier, different from the results of the KNN classifier, the performance of all the algorithms starts to be stable until the size of selected bands reaches 20. When the size of selected bands is located in the range of 20–30, ONRBS outperforms other algorithms. When the size of selected bands is more than 30, our proposed SNEA can reach the highest accuracy. Under the metric of AA, our proposed SNEA obviously has a leading advantage over ONRBS and LvaHAI, because the Indian_Pines dataset has 16 classes of ground objects, and the numbers of samples of different classes are imbalanced, for example, *Oats* and *Stone-Steel-Tower* contain a small number of samples, and there may be low classification accuracy in these classes for other algorithms. Through the optimization on the local consistency with adjacent neighboring pixels and nonlocal consistency among pixels, our proposed SNEA cannot be affected by the imbalanced samples.

The AA of the selected bands with different band numbers (10–60) obtained by all the algorithms is shown in Table II. We can notice that the AA of all the band subsets obtained by our proposed SNEA is the highest for both the KNN and SVM classifiers. In detail, for the KNN classifier, the average OA, Kappa, and AA of our proposed SNEA can reach 0.7285, 0.6882, and 0.6249, which significantly outperform those of other algorithms. Note that when BS is not utilized and the features of all bands are applied to the KNN classifier, the OA, Kappa, and AA are 0.6820, 0.6341, and 0.5597. After BS, the distinct bands for classification are selected out, and the noisy bands, which are even reducing the classification, are excluded. For the SVM classifier, the average OA, Kappa, and AA of all the band subsets obtained by our proposed SNEA can reach 0.8021, 0.7737, and 0.8095, which outperform those of other algorithms. Compared with the results of all bands, the OA and Kappa are a little worse, but the AA is obviously better.

2) Salinas: Similar to the Indian_Pines dataset, the OA, Kappa, and AA results of the KNN and SVM classifiers on the Salinas dataset are illustrated in Fig. 9. For the KNN classifier, it can be noticed that MOSCA, LvaHAI, ONRBS, and our proposed SNEA are much better than other algorithms, and meanwhile, they all perform better than all bands. Note that in the whole range of selected band sizes, the performance of ONRBS and our proposed SNEA is very stable, while MOSCA may fluctuate at some selected band sizes. For the SVM classifier, the classification performances of all the algorithms are similar to the whole range of selected band sizes and are a little worse than that of all bands. It can also be observed that at the end of small size of selected band subset, there is an obvious variation, and when the size of

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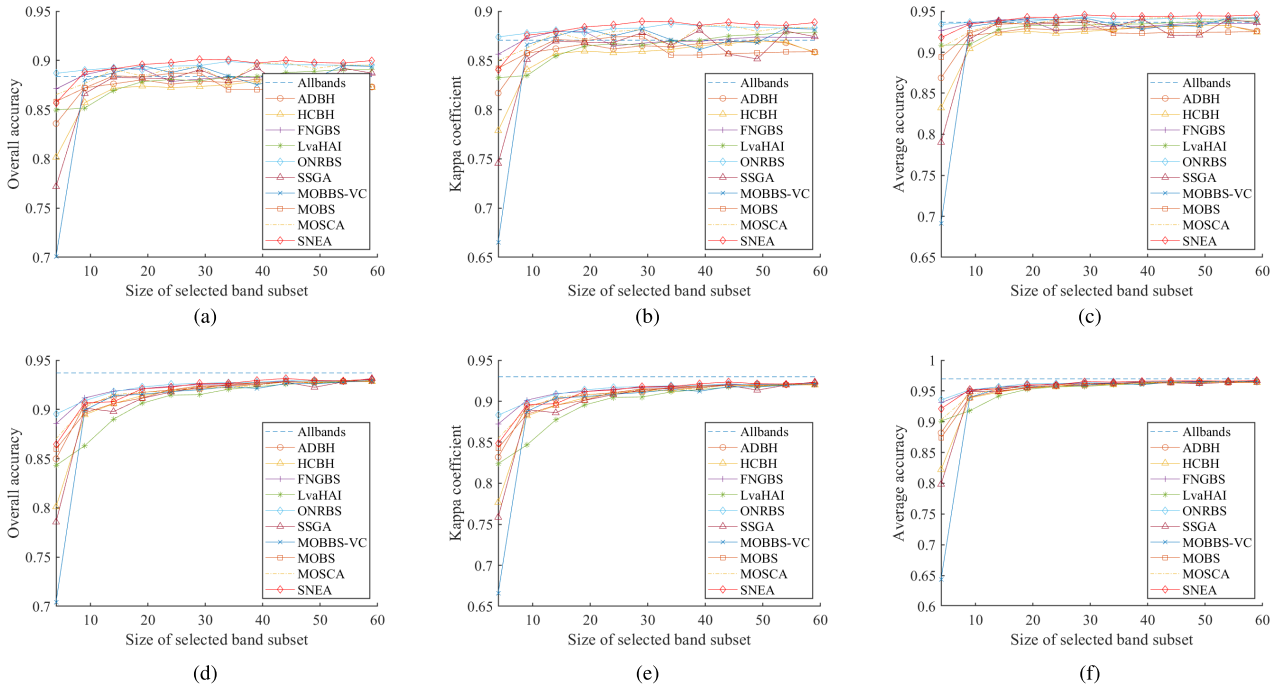


Fig. 9. OA, Kappa, and AA results of all the algorithms on the Salinas dataset with the KNN classifier and the SVM classifier. (a) KNN-OA. (b) KNN-Kappa. (c) KNN-AA. (d) SVM-OA. (e) SVM-Kappa. (f) SVM-AA.

TABLE II
AVERAGE OA, KAPPA, AND AA RESULTS OF 10–60 SELECTED BANDS ON THE INDIAN_PINES DATASET

Indian_Pines(10-60 bands)						
	KNN			SVM		
	OA	Kappa	AA	OA	Kappa	AA
Allbands	0.6820	0.6341	0.5597	0.8025	0.7748	0.7219
ADBH	0.6295	0.5729	0.5017	0.7648	0.7316	0.7045
HCBH	0.6632	0.6124	0.5432	0.7812	0.7502	0.7328
FNGBS	0.6932	0.6470	0.5727	0.7860	0.7555	0.7554
LvaHAI	0.7001	0.6557	0.5691	0.7953	0.7654	0.7726
ONRBS	0.6894	0.6430	0.5681	0.7953	0.7664	0.7607
SSGA	0.6476	0.5940	0.5249	0.7608	0.7263	0.7133
MBBS-VC	0.6566	0.6046	0.5341	0.7748	0.7426	0.7339
MOBS	0.6521	0.5996	0.5280	0.7661	0.7327	0.7168
MOSCA	0.6566	0.6049	0.5324	0.7703	0.7375	0.7226
SNEA	**0.7285**	**0.6882**	**0.6249**	**0.8021**	**0.7737**	**0.8095**

TABLE III
AVERAGE OA, KAPPA, AND AA RESULTS OF 10–60 SELECTED BANDS ON THE SALINAS DATASET

Salinas(10-60 bands)						
	KNN			SVM		
	OA	Kappa	AA	OA	Kappa	AA
Allbands	0.8849	0.8717	0.9367	0.9367	0.9294	0.9689
ADBH	0.8784	0.8645	0.9284	0.9235	0.9146	0.9620
HCBH	0.8763	0.8621	0.9273	0.9211	0.9119	0.9593
FNGBS	0.8851	0.8719	0.9336	0.9255	0.9169	0.9625
LvaHAI	0.8824	0.8690	0.9339	0.9158	0.9060	0.9578
ONRBS	0.8947	0.8828	0.9401	0.9257	0.9170	0.9626
SSGA	0.8805	0.8668	0.9313	0.9199	0.9106	0.9597
MBBS-VC	0.8874	0.8745	0.9362	0.9214	0.9122	0.9605
MOBS	0.8787	0.8649	0.9295	0.9224	0.9134	0.9599
MOSCA	0.8893	0.8766	0.9363	0.9235	0.9147	0.9611
SNEA	**0.8964**	**0.8846**	**0.9436**	**0.9258**	**0.9171**	**0.9638**

selected band subset is increased, the variation decreases a lot. This means all the algorithms can select the effective bands at these selected band sizes.

The AA of the selected bands with different selected band sizes (10–60) obtained by all the algorithms is shown in Table III. We can notice that the AA of our proposed SNEA is the highest for both KNN and SVM classifiers. In detail, for KNN classifier, the average OA, Kappa, and AA of our proposed SNEA can reach 0.8964, 0.8846, and 0.9436, which ranks the first in all the algorithms. Note that the OA, Kappa, and AA of all bands are 0.8849, 0.8717, and 0.9367, which are inferior to our proposed SNEA. Therefore, our proposed SNEA can select the distinct bands for classification and remove the noisy bands of the Salinas dataset. For the SVM classifier, the average OA, Kappa, and AA of SNEA of our proposed SNEA can reach 0.9258, 0.9171, and 0.9638, which outperform all other algorithms but are a little inferior to those of all bands. Though the selected bands are only a fraction of all bands and the data amount is reduced greatly, the accuracy is decreased lightly.

3) PaviaU: Similar to the Indian_Pines and Salinas datasets, the OA, Kappa, and AA results of the KNN and SVM classifiers on the PaviaU dataset are illustrated in Fig. 10. For the KNN classifier, it can be noticed that LvaHAI, ONRBS, and our proposed SNEA are much better than other algorithms, and they all perform better than all bands. Note that at the end of small size of selected band subset, the classification performance of our proposed SNEA is much better than ONRBS and LvaHAI, but when the band number exceeds 20, the classification performance of ONRBS and LvaHAI performs better than our proposed SNEA. When the selected band size is more than 40, the performance of our proposed SNEA surpasses ONRBS and LvaHAI again. For the SVM classifier, LvaHAI, ONRBS, FNGBS, and our proposed SNEA

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TABLE IV
AVERAGE OA, KAPPA, AND AA RESULTS OF 10–60 SELECTED BANDS ON THE PAVIAU DATASET

PaviaU(10-60 bands)						
	KNN			SVM		
	OA	Kappa	AA	OA	Kappa	AA
Allbands	0.8575	0.8065	0.8233	0.9381	0.9177	0.9182
ADBH	0.8480	0.7935	0.8099	**0.9238**	**0.8982**	**0.8961**
HCBH	0.8401	0.7826	0.8002	0.9202	0.8933	0.8912
FNGBS	0.8600	0.8101	0.8296	0.9162	0.8875	0.8877
LvaHAI	0.8787	0.8362	0.8482	0.9124	0.8828	0.8606
ONRBS	0.8768	0.8333	0.8473	0.9180	0.8901	0.8819
SSGA	0.8558	0.8044	0.8199	0.9127	0.8829	0.8737
MBBS-VC	0.8604	0.8108	0.8306	0.9049	0.8719	0.8728
MOBS	0.8427	0.7862	0.8059	0.9131	0.8837	0.8825
MOSCA	0.8601	0.8101	0.8277	0.9149	0.8859	0.8847
SNEA	**0.8794**	**0.8371**	**0.8490**	0.9132	0.8839	0.8741

are much better than other algorithms, and they all perform worse than all bands. When the size of selected band subset is small, the performance of our proposed SNEA is the best. But when the size of selected band subset is increased (more than 20), the performance of our proposed SNEA has an obvious degradation, and FNGBS starts to rank the first in all the algorithms. When the size of selected subset reaches 40, LvaHAI, ONRBS, FNGBS, and our proposed SNEA keep almost the same pace to approximate the accuracy of all bands.

The AA of the selected bands with different selected band sizes (10–60) obtained by all the algorithms is shown in Table IV. We can see that our AA is the highest for the KNN classifier, and ADBH is the best for the SVM classifier. In detail, for the KNN classifier, the average OA, Kappa, and AA of our proposed SNEA can reach 0.8794, 0.8371, and 0.8490, which ranks the first in all the algorithms, including all bands whose OA, Kappa, and AA are 0.8575, 0.8065, and 0.8233. Our proposed SNEA is effective in selecting appropriate band subset for the KNN classifier. For the SVM classifier, the average OA, Kappa, and AA of our proposed SNEA can reach 0.9132, 0.8839, and 0.8741, which are inferior to ADBH, ONRBS, MOSCA, and all bands.

On the whole, our proposed SNEA can achieve better band subsets than the comparison algorithms. The superiority on the KNN classifier is more obvious than the SVM classifier. The classification results of the three whole HSIs when the band number is set to 30 and KNN is taken as the classifier are illustrated in Supplementary Materials. From these results, we can notice that for the regions with the same ground object, the band subset obtained through our proposed SNEA can reflect the distinct features better, and there are much less samples, which are wrongly classified. The reason is attributed to two aspects: for one thing, the local consistency optimization objective directly aims at reducing the inconsistency at feature level, and it is beneficial for the selected bands to the classification. For another, the neighborhood-grouping strategy can choose the bands, which are distributed more evenly in the band space and are more robust to noise.

C. Effectiveness of the Proposed Strategies in SNEA

The prominent accuracy of our proposed SNEA is attributed to the proposed strategies, which include the local/nonlocal consistency objectives and neighborhood-grouping-based pairwise learning strategy. To verify the effectiveness of the proposed strategies, first, two schemes are designed based on SNEA: one is to verify the effectiveness of the local/nonlocal consistency objectives and is named SNEA-MIVar, where two common objectives (the band mean variance (Var) and the MI among bands) are adopted, and the remaining parts are kept the same with SNEA. The other one is to verify the effectiveness of neighborhood-grouping-based pairwise learning strategy and is named SNEA-WOG, where group-based population initialization is removed and replaced by random initialization, and importance-based pairwise learning is removed and replaced by a random single-point crossover operation. Second, to further investigate the effectiveness of the components in the neighborhood-grouping-based pairwise learning strategy, two variant algorithms are further designed based on SNEA: the first one is constructed by only replacing the neighborhood grouping with the random grouping, namely SNEA-RandG. The second one is constructed by replacing importance-based pairwise learning with random pairwise learning in the group where $p(i)$ is equal to 0.5, namely, SNEA-RandL.

The experimental results of OA on the three datasets are shown in Fig. 11. It can be observed that the results of SNEA-MIVar and SNEA-WOG are significantly lower than SNEA. Meanwhile, SNEA-MIVar is also inferior to SNEA-WOG on the Indian_Pines dataset and ties with SNEA-WOG on the Salinas dataset, but performs better than SNEA-WOG on the PaviaU dataset. In detail, the OA of SNEA-WOG fluctuates greatly when the number of bands is relatively high, especially on the Indian_Pines dataset. The reason lies on that a lot of redundant bands will be selected in the situation of large size of selected band subset, and the grouping strategy can effectively avoid selecting too many bands with the same discriminating capability at the same time and make the selected bands evenly distributed. Therefore, the local/nonlocal consistency objectives and neighborhood-grouping-based pairwise learning strategy are both effective to improve the performance of our proposed SNEA. From Fig. 11, it can also be observed that SNEA-RandG and SNEA-RandL are both inferior to SNEA. The neighborhood-grouping and importance-based pairwise learning are both effective for the performance of the strategy of neighborhood-grouped pairwise learning.

Besides, as our proposed SNEA is an MOEA to optimize the maintenance of local structure and global structure simultaneously, at the end of evolutionary process, a serial of optimal solutions (PF), which are nondominated to each other, are obtained. However, only one band subset is used for the classification task, and a solution strategy should be adopted to select one subset from the PF. In this experiment, the metrics to be considered include the following: 1) f_1 and 2) f_2, which are the optimization objectives in SNEA; 3) the SDE; 4) the MI of the bands among selected bands; and 5) the average of the variances of the selected bands (Var). Obviously, the band subset is better when the metrics of f_1, f_2, MI, and SDE are lower. The band subsets obtained through SNEA are selected according to these five metrics, and the detailed results

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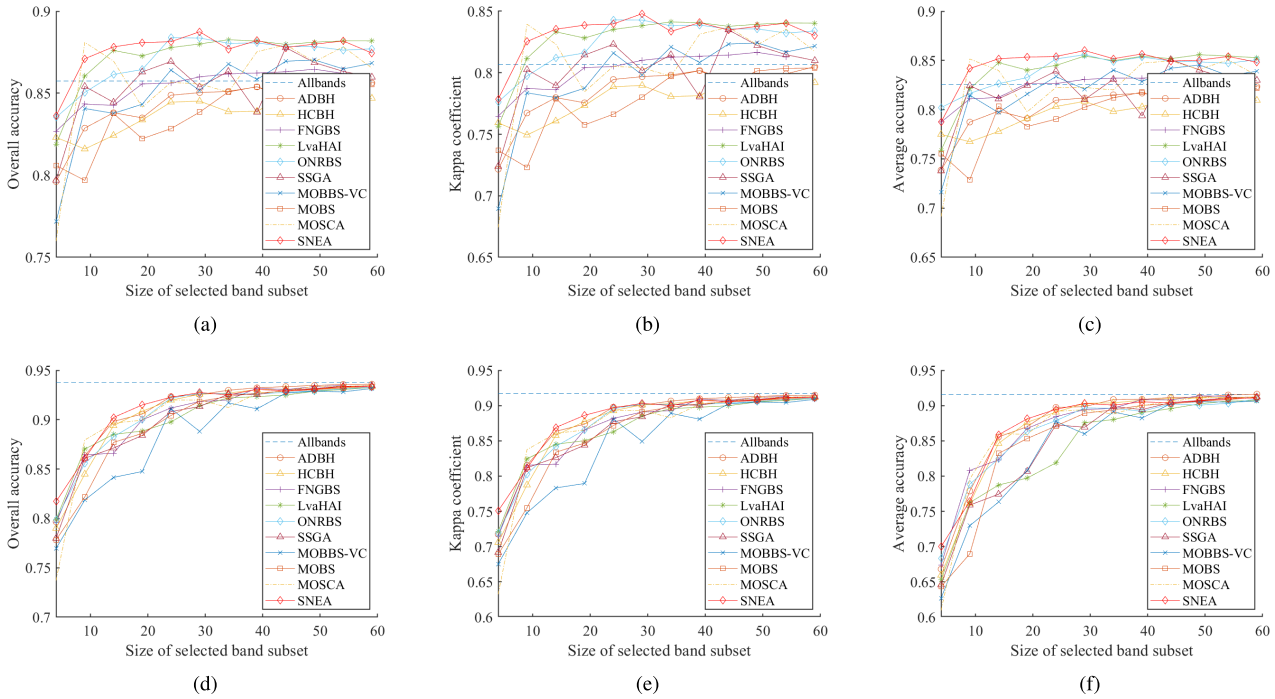


Fig. 10. OA, Kappa, and AA results of all the algorithms on the PaviaU dataset with the KNN classifier and the SVM classifier. (a) KNN-OA. (b) KNN-Kappa. (c) KNN-AA. (d) SVM-OA. (e) SVM-Kappa. (f) SVM-AA.

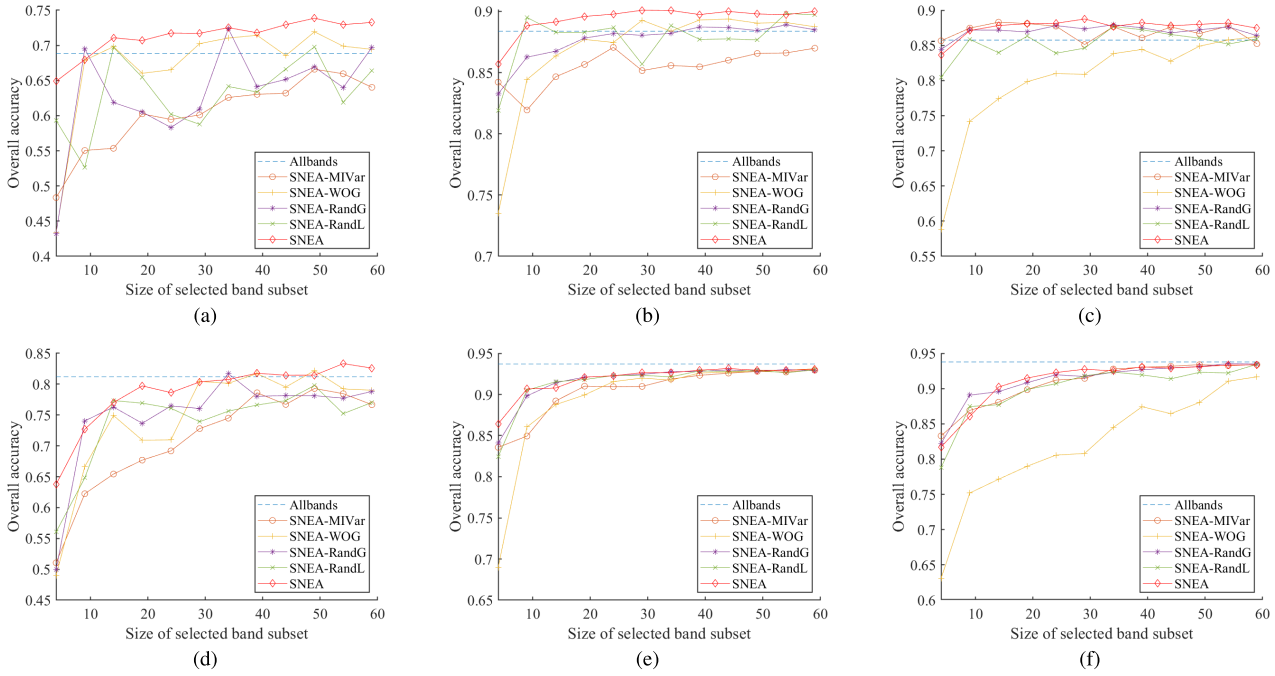


Fig. 11. OA results on the Indian_Pines, Salinas, and PaviaU datasets with the KNN classifier and the SVM classifier for SNEA and the four SNEA variants. (a) Indian_Pines-KNN-OA. (b) Salinas-KNN-OA. (c) PaviaU-KNN-OA. (d) Indian_Pines-SVM-OA. (e) Salinas-SVM-OA. (f) PaviaU-SVM-OA.

are shown in Table V. Note that the best accuracy obtained from the selected band subsets in PF is shown in the first row (max). It can be observed that on the Indian_Pines and Salinas datasets, the band subset selected by the f_2 metric can achieve the best classification accuracy on the Indian_Pines dataset and the best AA on the Salinas dataset, while on the PaviaU dataset, the band subset determined by the metric of MI can achieve the best classification accuracy. In all of our experiments, the f_2 metric is used as the solution selection metric.

D. Parameter Sensitivity Analysis

In our proposed SNEA, there is a crucial parameter, which is the number of nearest neighbors K_g in the affinity graph.

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TABLE V

CLASSIFICATION ACCURACY OF THE SELECTED BANDS FROM DIFFERENT SELECTION STRATEGIES WITH THE KNN CLASSIFIER (AVERAGE OF 10–60 SELECTED BANDS)

	Indian_Pines			Salinas			PaviaU		
	OA	Kappa	AA	OA	Kappa	AA	OA	Kappa	AA
Max	0.7215	0.6809	0.6041	0.9000	0.8887	0.9442	0.8804	0.8385	0.8519
$f1$	0.6313	0.5746	0.5022	0.8926	0.8804	0.9358	0.8758	0.8321	**0.8452**
$f2$	**0.7120**	**0.6698**	**0.5668**	0.8957	0.8838	**0.9424**	0.8642	0.8163	0.8312
MI	0.6880	0.6417	0.5475	0.8943	0.8823	0.9379	**0.8759**	**0.8324**	0.8441
Var	0.6407	0.5856	0.5086	**0.8965**	**0.8847**	0.9420	0.8689	0.8226	0.8381
SDE	0.6707	0.6212	0.5364	0.8940	0.8819	0.9387	0.8701	0.8244	0.8370

TABLE VI

AVERAGE OA, KAPPA, AND AA RESULTS OF 10–60 SELECTED BANDS (DERIVED BY SNEA WITH DIFFERENT VALUES OF K_g) ON INDIAN_PINES, SALINAS, AND PAVIAU WITH THE KNN AND SVM CLASSIFIERS

Indian_Pines						
	KNN			SVM		
	OA	Kappa	AA	OA	Kappa	AA
$K_g = n/2$	0.7074	0.6641	0.5803	0.7781	0.7454	0.7524
$K_g = n/4$	0.7019	0.6578	0.5806	0.7756	0.7422	0.7445
$K_g = n/8$	**0.7285**	**0.6882**	**0.6249**	**0.8021**	**0.7737**	**0.8095**
$K_g = n/16$	0.7043	0.6605	0.5780	0.7767	0.7437	0.7443
Salinas						
	KNN			SVM		
	OA	Kappa	AA	OA	Kappa	AA
$K_g = n/2$	0.8850	0.8719	0.9351	0.9160	0.9061	0.9560
$K_g = n/4$	0.8853	0.8722	0.9353	0.9150	0.9051	0.9556
$K_g = n/8$	**0.8964**	**0.8846**	**0.9436**	**0.9258**	**0.9171**	**0.9638**
$K_g = n/16$	0.8864	0.8735	0.9357	0.9171	0.9074	0.9566
PaviaU						
	KNN			SVM		
	OA	Kappa	AA	OA	Kappa	AA
$K_g = n/2$	0.8515	0.7985	0.8174	0.8883	0.8484	0.8463
$K_g = n/4$	0.8631	0.8147	0.8308	0.9015	0.8672	0.8597
$K_g = n/8$	**0.8794**	**0.8371**	**0.8490**	**0.9132**	**0.8839**	**0.8741**
$K_g = n/16$	0.8547	0.8029	0.8226	0.8920	0.8534	0.8521

In this experiment, K_g is set to $n/16$, $n/8$, $n/4$, and $n/2$, respectively, where n is the number of pixels in EvalSet. The detailed results are shown in Table VI. It can be observed that when K_g is equal to $n/8$, the best classification performance on all the datasets can be achieved. It can be explained as follows: for one thing, if K_g is set to small values, such as $n/16$, some key relationship may be neglected, and the affinity graph cannot reflect most of the relationship among samples. For another, if K_g is set to large values, such as $n/2$, some noisy relationship may be introduced into the affinity graph. Therefore, the high-quality selected band subset cannot be obtained. In all of our experiments, K_g is set to $n/8$.

E. Results on the Houston Dataset

The Houston dataset was released in the IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest in 2013[2] and covers the University of Houston campus. The dataset consists of 15 classes and includes 144 spectral bands with a spatial size of 349 × 1905 and a spatial resolution of 2.5 m per pixel. The challenge for the Houston dataset lies on the large pixel number (about 700 000), and thus, the MI or Var metrics are difficult to achieve for some hyperspectral BS algorithms. In order to validate the effectiveness of our proposed SNEA, on the Houston dataset, the classification results of all the algorithms are collected and shown in Table VII. Due to the large image size, the algorithms MOSCA, MOBS, and MOBBS-VC cannot obtain the whole band subsets in 24 h. From Table VII, it can be observed that our proposed SNEA can achieve the best performance with both the KNN and SVM classifiers on the Houston dataset. The average OA, AA, and Kappa of SNEA can reach 0.8608, 0.8493, and 0.8430 with the KNN classifier and 0.8713, 0.8607, and 0.8654 with the SVM classifier. These results consist of the results on the three datasets (Indian_Pines, PaviaU, and Salinas). It also reveals that our proposed SNEA can handle the HSI with a large image size. The OA, Kappa, and AA results of KNN and SVM classifiers on the Houston dataset with different numbers of selected bands are illustrated in Fig. 12.

TABLE VII

AVERAGE OA, KAPPA, AND AA RESULTS OF 10–60 SELECTED BANDS ON THE HOUSTON DATASET

Houston(10-60 bands)						
	KNN			SVM		
	OA	Kappa	AA	OA	Kappa	AA
Allbands	0.8618	0.8504	0.8436	0.9208	0.9143	0.9165
ADBH	0.8504	0.8380	0.8333	0.8640	0.8528	0.8557
HCBH	0.8549	0.8429	0.8370	0.8681	0.8572	0.8605
FNGBS	0.8580	0.8463	0.8407	0.8641	0.8529	0.8577
LvaHai	0.8464	0.8337	0.8291	0.8520	0.8398	0.8432
ONRBS	0.8503	0.8379	0.8326	0.8583	0.8466	0.8517
SSGA	\	\	\	\	\	\
MBBS-VC	\	\	\	\	\	\
MOBS	\	\	\	\	\	\
MOSCA	0.8436	0.8307	0.8269	0.8473	0.8347	0.8395
SNEA	**0.8608**	**0.8493**	**0.8434**	**0.8713**	**0.8607**	**0.8654**

F. Results With the CNN-Based HSI Classification Model

In recent years, with the hierarchical and representation learning characteristics, deep learning technologies have drawn the increasing attention of the literature. Unlike traditional methods, such as SVM, which receives the designed handcrafted features, the deep neural networks (DNNs) can learn representative features automatically. In this experiment, a widely used deep learning model for HSI classification, namely, adaptive spectral spatial multiscale network for hyperspectral image classification (ASSMN) [50], is adopted to evaluate the performance of the band subsets obtained from our proposed SNEA. ASSMN is an end-to-end adaptive spectral–spatial multiscale network to extract multiscale contextual information for HSI classification, which contains spectral FE and spatial FE subnetworks. Normally, the all-bands

[2]https://hyperspectral.ee.uh.edu/?page_id=459

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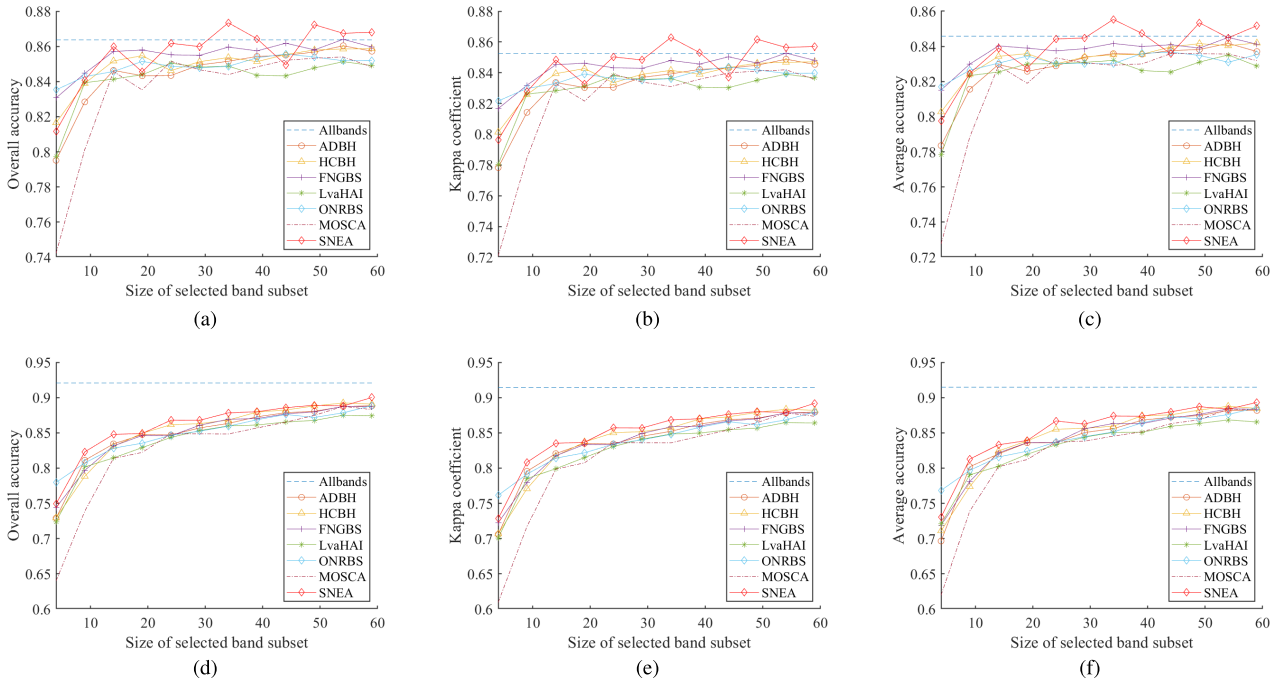


Fig. 12. OA, Kappa, and AA results of all the algorithms on the Houston dataset with the KNN classifier and the SVM classifier. (a) KNN-OA. (b) KNN-Kappa. (c) KNN-AA. (d) SVM-OA. (e) SVM-Kappa. (f) SVM-AA.

TABLE VIII
CLASSIFICATION RESULTS OF THE CNN-BASED MODEL ASSMN WITH THE SELECTED BANDS OF SNEA

	Indian_Pines			PaviaU			Salinas		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
SNEA(30 bands)	0.9677	0.9824	0.9628	0.9495	0.9729	0.9341	**0.9819**	**0.9928**	**0.9799**
SNEA(60 bands)	**0.9862**	0.9675	**0.9840**	**0.9820**	**0.9856**	**0.9761**	0.9817	0.9924	0.9796
Allbands	0.9749	**0.9889**	0.9710	0.9692	0.9711	0.9592	0.9730	0.9893	0.9700

TABLE IX
RUNTIME TIME FOR OUR PROPOSED SNEA AND SEVEN COMPARISON ALGORITHMS FOR THE SELECTION OF 10–60 BANDS ON INDIAN_PINES, SALINAS, AND PAVIAU DATASETS (S)

	FNGBS	LvaHAI	ONRBS	SSGA	MOBBS-VC	MOBS	MOSCA	SNEA
Indian_Pines	**0.2065**	31.9597	37.1821	1993.5700	4859.3149	651.9400	2359.3100	198.7400
Salinas	**0.3459**	621.0236	42.1769	3941.8300	22374.9900	1655.6900	7376.9900	1024.6700
PaviaU	**0.0740**	53.7745	16.0262	4744.7000	71436.3000	3059.2300	10236.3000	623.4500
Average	**0.2088**	235.5859	31.7951	3560.0333	32890.2016	1788.9533	6657.5333	615.6200

information of pixels is fed into ASSMN. To demonstrate the effectiveness of our proposed SNEA, only the information of the selected bands for each pixel is taken as the input for ASSMN. For each band subset, ASSMN is retrained and tested on the same training dataset with the original ASSMN model whose input data are the all-bands information. Considering the high complexity of training the ASSMN model for different band subsets, the band subsets with the number of 30 and 60 bands from our proposed SNEA are taken as the evaluation points, and the detailed results are shown in Table VIII. From these results, we can observe that though the number of bands are decreased to only a fraction of all bands, the classification performance of ASSMN is kept the same or superior to those of all bands. The SNEA algorithm can extract effective bands from the whole band space and can help improve the performance of the CNN-based HSI classification model.

G. Running Time

The running time of all the algorithms is collected in Table IX. On the whole, EA-based algorithms (SSGA, MOBBS-VC, MOBS, MOSCA, and SNEA) consume much more time than non-EA-based algorithms (FNGBS, LvaHAI, and ONRBS), since EAs refer to a lot of fitness evaluations. The non-EA-based methods only need to select a fixed subset of bands according to predefined rules, but the classification performance is inferior to our proposed SNEA. As for the EA-based algorithms, our proposed SNEA is much faster than SSGA, MOBBS-VC, MOBS, and MOSCA. The reason lies on two aspects: for one thing, the neighborhood-grouping-based pairwise learning strategy can maintain an even distribution in the band space and accelerate the convergence of SNEA. For another, in SNEA, only a small part of the whole pixels is used to construct the EvalSet and further to calculate the objectives, avoiding using the whole pixels to evaluate the fitness in many

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algorithms, such as MOBS, MOSCA, and so on. Combining the accuracy and running time, our proposed SNEA has a good balance between accuracy and time complexity.

V. Conclusion

In this work, an unsupervised hyperspectral BS method through SNEA is proposed. Different from other EAs for hyperspectral BS, first, two spatial-structure-related optimization objectives are designed, including the locally spatial structure denoted by the pixel's spatial consistency with its adjacent neighbors and the globally spatial structure denoted by the affinity graph among pixels. The hyperspectral BS is formulated as the problem of conserving spatial structures. Moreover, a neighborhood-grouped pairwise learning strategy is proposed to generate high-quality offsprings. In this novel strategy, a neighborhood-grouping operation is designed to divide the band space into several groups. The population can be initialized efficiently, and the offspring solutions can be generated pairwisely under the guidance of grouping. Compared with nine state-of-the-art comparison algorithms, the experimental results on three standard hyperspectral datasets demonstrate that the band subset obtained by our proposed SNEA has a better classification performance than the comparison algorithms.

In future, the reduction of computational complexity for our proposed SNEA will be further investigated, and some efficient surrogate models will be designed. However, for the dimension of HSIs, only from the aspects of information entropy, correlation, and spatial structure, it is still not enough to get the best band subset for classification without the participation of classifiers. Therefore, in future work, hyperspectral BS using the wrapper-based method will be investigated, and meanwhile, the high-computational burden brought by the wrapper-based hyperspectral BS should be decreased to an acceptable extent.

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